Ch-4 → Number theory & cryptography

(*) **Def^n 1:** If $a, b \in Z$, $a \neq 0$, $c \in Z$ such that $b = ac$

then "a divides b"

↳ b/a or $a \mid b$ → b div a

divided by

$a = 3$, $b = 7$, $3 \nmid 7$	$a = 12$, $b = 24$, $12 \mid 24$

(*) **Def^n 2:**

→ divisor (d) | dividend (a) | quotient (q)

remainder (r)

$q = a \text{ div } d = a/d$

$r = a \bmod d$

$a = d.q + r$, $0 \leq r < d$

$21/5 \to 4....$ $\quad 0 \leq r < d$

$21 \% 5 \to 1$

* **Defn 3** : If $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$ then

→ $a \equiv b \pmod{m}$ if $m \mid (a-b)$ ↪ $\frac{a-b}{m}$

o/w, $a \not\equiv b \pmod{m}$

eg: → $101 \equiv 2 \bmod 11$ → $(101-2) \to 99$

↪ $m = 11$ $\quad 11 \mid 99$

$100 \equiv 2 \bmod 11$ → $(100-2) \to 98/11$ ✗

* **Theorem - 3** : $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$ then

$a \equiv b \pmod{m}$ iff $a \pmod{m} = b \pmod{m}$

↪ a, b have the same remainder on division by m.

Theorem-4 let $m \in \mathbb{Z}^+$, a, b are congruent modulo m if there is an integer k such that

$$a = b + km$$

$\rightarrow b/a = c \rightarrow b = ac + 0$

$a = d \cdot q + r$

a/b

Derivation

$a \equiv b \pmod m$

$\rightarrow m \mid (a-b) \longrightarrow$ defn 3

$\rightarrow (a-b) = km \longrightarrow$ defn 1

$\rightarrow a = b + km$

Theorem - 5 : let,

$m \in \mathbb{Z}^+$

$a \equiv b \pmod m$

$c \equiv d \pmod m$

$\rightarrow$ (1) $(a+c) \equiv (b+d) \pmod m$

(2) $ac \equiv (bd) \pmod m$

Corollary: let $m \in \mathbb{Z}^+$ and $a, b \in \mathbb{Z}$ then,

$\hookrightarrow$ (1) $(a+b) \bmod m = \big((a \bmod m) + (b \bmod m)\big) (\bmod m)$

(2) $(ab) \bmod m = \big((a \bmod m)(b \bmod m)\big) (\bmod m)$

$\boxed{\text{Arithmatic modulo } m}$

$\mathbb{Z}_m \in$ set of $\mathbb{Z}^- < m$

$\hookrightarrow$ $\underline{\mathbb{Z}_m} \in \mathbb{Z}^-$ such $\mathbb{Z}_m < m$ then

→ Addition $a +_m b \equiv (a+b) \bmod m$

→ Multiplication $a \cdot_m b \equiv (ab) \bmod m$

For example:

$7 +_{11} 9 \equiv (7+9) \bmod 11 \rightarrow 16 \bmod 11 \rightarrow 5$

$7 \cdot_{11} 9 \equiv (7 \cdot 9) \bmod 11 \rightarrow 63 \bmod 11 \rightarrow 8$

Corollary: let $m \in \mathbb{Z}^+$ and $a, b \in \mathbb{Z}$ then,

1. $(a+b) \bmod m = ((a \bmod m) + (b \bmod m)) \pmod m$
2. $(ab) \bmod m = ((a \bmod m)(b \bmod m)) \pmod m$

$$\boxed{\text{Arithmatic modulo } m}$$

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